



(e) The air-conditioning system uses a temperature sensor and a microprocessor to keep the temperature of the room at a constant level.

(i) Tick (✓) **one** box to show whether the sensor is an example of an input, output, process or storage device.

- |                  |                          |
|------------------|--------------------------|
| <b>A</b> input   | <input type="checkbox"/> |
| <b>B</b> output  | <input type="checkbox"/> |
| <b>C</b> process | <input type="checkbox"/> |
| <b>D</b> storage | <input type="checkbox"/> |

[1]

(ii) The temperature sensor continuously sends analogue data to the microprocessor. This data is then converted to digital.

Explain how the microprocessor keeps the temperature of the room at a constant level.

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..... [3]

(f) The air-conditioning system is an example of an embedded system.

Explain why the air-conditioning system is an embedded system.

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..... [2]

